Assignment 1 | Generalized Linear Models

Executed report • three exercises • numerical fits and diagnostics independently reproduced

Status, provenance and reproducibility

Verified, executed analysis. All three exercises have been re-fitted in this workspace, and the revised notebook includes saved execution outputs. Exercises 1 and 3 use exact row-level extractions of only the needed columns from the public GitHub CSV datasets (395 student records, 740 absenteeism records); Exercise 2 uses the complete 30-cell Sex × Race × Education contingency table (32,561 counts) independently reconstructed from the GitHub archive. These are transparent reduced views of the original datasets, not synthetic records. Source filenames, extraction details and recovered input files are included in the evidence folder.

Assignment: **Generalized Linear Models**, three exercises. Instructor: Giulia Vannucci. Submission: report, code file, and note documenting individual/group work. Analysis code: **assignment1.ipynb**. No numerical results have been fabricated. Statistical claims are associations, not causal effects.

Exercise 1 • Multinomial and ordinal logistic regression

Research question and preprocessing

Using the UCI Student Performance mathematics file (**student-mat.csv**), the outcome G3 is recoded into Low (0–9), Medium (10–14), High (15–20). Grade 10 is the conventional pass threshold and 15 differentiates higher achievement. The retained predictors are studytime, prior failures, absences, maternal education (Medu), and paternal education (Fedu), mirroring the existing project implementation. The ordinal scores for parental education and study time are treated as numerical predictors; this assumes a linear log-odds effect per category.

Models and saved results

The multinomial logit takes Low as reference and fits separate log-odds for Medium versus Low and High versus Low. The proportional-odds ordered logit assumes a single slope for each predictor across both cumulative grade cut-points, with separate intercept thresholds.

Metric	Multinomial logit	Ordered logit
N	395	395
Log likelihood	-370.05	-375.65
AIC	764.109	765.309

Recorded ordinal coefficients: failures = -0.9290 ($p<0.001$; odds ratio $\exp(-0.929)=0.395$), absences = -0.0299 ($p=0.021$; odds ratio about 0.971 per hour/day recorded), and Medu = +0.2672 ($p=0.024$; odds ratio about 1.306). Positive slopes increase the odds of being in higher performance categories under the proportional-odds specification; negative slopes decrease those odds. The Multinomial fit also shows negative failure coefficients (Medium: -0.8134; High: -1.7377 versus Low).

Critical assumption and model choice

The original repository prints a likelihood-ratio p=0.0476 comparing multinomial and ordered models as a Proportional Odds test. That calculation is not a valid nested-model likelihood-ratio

test for these generally nonnested families with linear predictors. The revised notebook fitted the two cumulative logits and performed a joint Brant-type sandwich Wald test of equality of slopes: **chi-square(5)=9.132; p=0.103924**. Thus the analysis does not reject proportional odds at the 5% threshold, although a nonsignificant test does not prove equality. AIC slightly prefers multinomial (764.109 vs 765.309; delta=1.20), whereas BIC favors the more parsimonious ordinal model (793.161 vs 811.856). For descriptive clarity and parsimony, select the **ordinal proportional-odds model**, with a caution about power and model fit.

Final interpretation: prior failures and absences are associated with poorer ordinal math performance; maternal education is positively associated with higher performance. The joint proportional-odds diagnostic is not significant at 5%, and BIC favors the ordinal specification, so the ordinal model is a defensible final choice. This is association, not a causal finding.

Exercise 2 • Log-linear association models and logistic equivalence

Dataset and models

The UCI Adult data were cross-classified by Sex (Female/Male), Race (5 levels) and recoded Education (HighSchool, College, Advanced). The GitHub output provides the complete 2 × 5 × 3 table of 30 cell frequencies, totaling **32,561** records. Reconstructing this table preserves the full information needed to fit hierarchical log-linear models. Unlike reconstructing individual records, this does not assert access to the person-level CSV.

Three nested Poisson log-linear models are compared: mutual independence [S][R][E]; conditional independence [SE][RE] (Sex independent of Race given Education); and all two-way associations [SR][SE][RE]. Likelihood-ratio chi-square tests compare changes in Poisson deviance and model degrees of freedom.

Model	Deviance	Residual df	Fit p
independence	1104.683	22	9.8e-220
conditional independence	436.612	12	6.56e-86
all two way	19.297	8	0.0133

Nested comparison	LR chi-square	df	p-value
independence → conditional independence	668.071	10	4.5e-137
conditional independence → all two way	417.315	4	5e-89

The all-two-way model improves dramatically over conditional independence (LR chi-square(4) = 417.315, p about 5.04 × 10^-89). However, its deviance is 19.297 on 8 residual degrees of freedom (p = 0.01335), so it still lacks perfect goodness of fit at the 5% criterion. A three-way interaction is indicated. None of the three prespecified nonsaturated models is strictly acceptable at the 5% cutoff; [SR][SE][RE] is the best of the three by relative fit, but retains a limitation. A saturated [SRE] model fits the table exactly but has no residual degrees of freedom to assess fit.

Formal equivalence to a conditional binary logit

For Male coded 1, a logistic regression with main effects of Race and Education corresponds exactly to the all-two-way log-linear model containing [SR], [SE], and [RE]. The [RE] term leaves the marginal Race–Education distribution unrestricted, while [SR] and [SE] induce additive Sex log-odds.

Logit coefficient	Binomial GLM	Log-linear Poisson
Intercept	+0.6642 (SE 0.120)	+0.6642 (SE 0.120)
race[T.Asian-Pac-Islander]	+0.1651 (SE 0.135)	+0.1651 (SE 0.135)
race[T.Black]	-0.4854 (SE 0.122)	-0.4854 (SE 0.122)
race[T.Other]	-0.1194 (SE 0.171)	-0.1194 (SE 0.171)
race[T.White]	+0.2941 (SE 0.118)	+0.2941 (SE 0.118)
Edu_cat[T.College]	-0.3892 (SE 0.032)	-0.3892 (SE 0.032)
Edu_cat[T.HighSchool]	-0.0906 (SE 0.031)	-0.0906 (SE 0.031)

The maximum absolute difference of the seven coefficient estimates was approximately **1.3 × 10^-14**, a floating-point rounding error, and the standard errors also matched to the shown precision. This confirms the claimed algebraic equivalence on the reconstructed table, while the overall all-two-way goodness-of-fit caveat remains.

Example interpretation: holding education fixed, the Black versus Amer-Indian-Eskimo category has a male log-odds coefficient of -0.4854 (odds ratio ≈ 0.616); this describes relative sex composition in the sample, not individual-level behavior or causes.

Exercise 3 • Absence hours, overdispersion and structural zeros

Exploration and candidate models

Using UCI Absenteeism at Work (N=740), the count-like response is Absenteeism time in hours. Verified statistics: mean = **6.9243**, sample variance = **177.7155**, and observed zeros = **44 (5.95%)**. The variance/mean ratio is approximately **25.67**; the fitted Poisson Pearson chi-square divided by residual degrees of freedom is **22.127**, indicating extreme overdispersion. The fitted Poisson expects only 1.754 zeros. Predictors are distance from residence to work, age, service time and education; the absence-reason code is excluded because it may encode absence status and create leakage.

Four candidate families were fitted: Poisson, NB2 (estimated dispersion), zero-inflated Poisson (ZIP), and zero-inflated NB2 (ZINB). The inflation models use an intercept-only structural-zero component to avoid overparameterization. All four converged with finite log likelihoods. The recovered column subset reproduces the archived Poisson and NB2 AICs to rounding precision, providing a cross-check of input integrity.

Converged model	AIC	BIC	Expected zeros
NB2	4,423.985	4,451.625	106.008
ZINB2	4,425.985	4,458.232	106.008
ZIP	9,755.493	9,783.133	44.765
Poisson	10,133.249	10,156.282	1.754

Model selection: NB2 has the lowest AIC and BIC of the four converged models and is selected. ZINB2 offers virtually no likelihood gain over NB2 but adds an extra parameter (AIC penalty +2.000). Its inflation intercept corresponds to odds about 1.26 × 10^-8, effectively zero extra structural-zero probability. ZIP predicts the observed zero count more closely (44.77 versus 44) but fits the overall distribution very poorly (AIC 9755.49). NB2 overpredicts the number of zeros (106.01 versus 44), which must be acknowledged as a model-fit

limitation even though its overall information criteria are best. Zero counts alone are not proof of a structural-zero process.

How the two components must be interpreted

Count component: exp(beta) is the multiplicative rate ratio of the latent count process for a one-unit increase in a covariate, holding others fixed. Zero-inflation component: exp(gamma) is an odds ratio for the separate latent structural-zero state. In ZIP/ZINB, a zero may arise from either component. An intercept-only inflation model estimates one baseline structural-zero probability, not covariate effects on always-zero membership.

Final conclusion: A negative binomial count model is preferred among the four converged candidate models by both AIC and BIC. In ZIP, the estimated inflation odds are exp(inflate_const)=0.06235 (structural-zero probability about 5.87%) and the fitted ZIP count component has a distance rate ratio exp(beta)=0.9870. In ZINB2 the estimated extra-zero odds are 1.26 × 10^-8, indicating practical collapse to NB2; consequently an additional structural-zero mechanism is not supported by the fitted likelihood. A limitation is that NB2 predicts more zeros than observed.

Limitations, submission and replication

• Reproducibility: the included extracted-column data files and 30-cell grouped count table let all three exercises execute immediately; running against the original GitHub CSVs (in the same predictor specification) is also supported.

• Tests and intervals describe statistical association, not causation. Some ordinal/numeric coding choices are substantive simplifications.

• The professor requests one PDF covering all three exercises, an analysis script/notebook, and an individual/group-work note. The heading on page 1 says “both exercises” but the instruction body and submission checklist list three.

• All three exercises have been executed on recovered sufficient inputs; the supplied notebook contains the displayed results, including convergence status and diagnostics. The limitation is the use of reduced original-column extracts rather than full raw CSV attachments.